

Lab 3, Act 3: Answer Key

From counts to a continuous measurement

NRES 746, Instructor Answer Key (not for student distribution)

Each item repeats the handout's question, then gives the answer, a short rationale, and mistakes to watch for. Questions with one correct answer are answered directly. Open-ended items (1d and the group discussion) get sample answers or discussion points instead. All numbers are computed in R when this file is knitted.

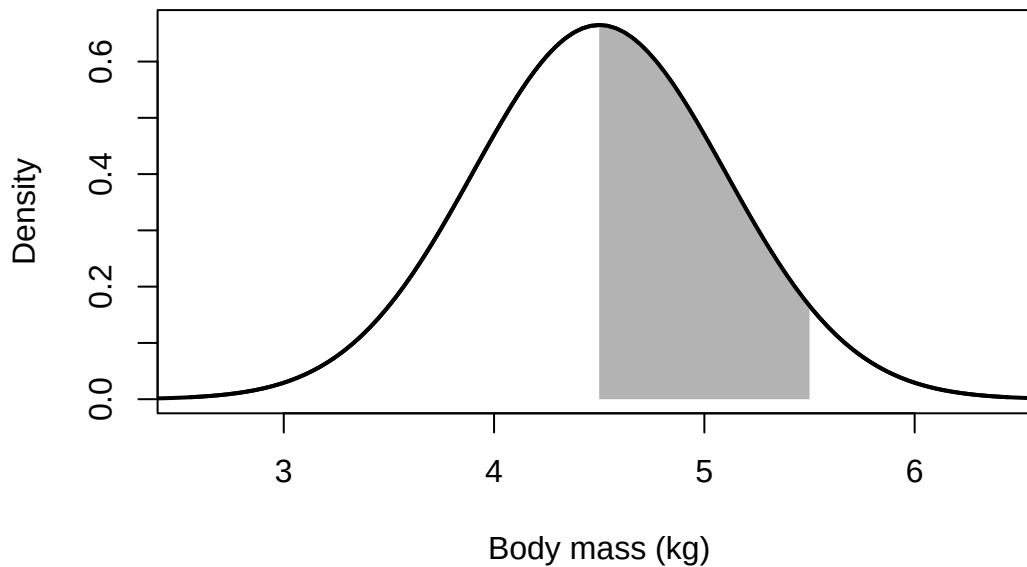
Setting: adult body mass is normal with mean $\mu = 4.5$ kg and standard deviation $\sigma = 0.6$ kg.

Part 1: Reading a density curve

1a.

Shade it by hand. On the curve above, shade the region that represents $P(4.5 < \text{mass} < 5.5)$.

Answer. The area under the curve between 4.5 and 5.5 kg, from the x-axis up to the curve:



Why. For a continuous variable, probability is area under the density over an interval. (This area is 0.4522.)

Watch for. Shading a thin strip at a single value, shading the space above the curve, or shading everything to the right of 4.5.

1b.

Integral notation. Given that the probability density function can be written $f(x)$, please write out the same region described above as a definite integral.

Answer.

$$P(4.5 < \text{mass} < 5.5) = \int_{4.5}^{5.5} f(x) dx$$

Why. A definite integral is the area under $f(x)$ between its limits, which is what 1a shaded.

Watch for. Missing dx , limits reversed, or writing $f(5.5) - f(4.5)$ (a difference of heights, not an area).

1c.

What is $P(\text{mass} = 4.5)$ exactly? Why? What's different about a single point on a continuous curve versus an interval?

Answer. 0. A single point has no width, so the area under the curve above it is zero, however tall the curve is there. Probability for a continuous variable only builds up over an interval.

Why. $\int_{4.5}^{4.5} f(x) dx = 0$.

Watch for. Reading the curve's height at 4.5 (about 0.66) as the probability. "Very small but not zero."

1d.

The height of the curve at $\text{mass} = 4.5$ is not zero (in fact, it's the highest point on the whole curve). If that height isn't a probability, what does it actually represent? Try to write out in words what a "probability density" means.

Sample answer. It's probability per unit of mass: how concentrated the probability is near 4.5 kg. Multiplying the height by a small width gives the approximate probability of landing in that small interval (e.g. $0.665 \times 0.1 \approx 0.07$ for 4.45 to 4.55 kg).

Why. Density has units of 1/kg here, which is why it only becomes a probability once multiplied by a width (or integrated over an interval).

Watch for. "The most likely value" with no explanation of what the height measures. Claims that a density can't exceed 1 (it can when σ is small).

Part 2: Reading a CDF

2a.

Using the CDF plot, estimate $P(\text{mass} > 5.5)$. You don't need to be precise, just in the right neighborhood.

Answer. About 0.05. The CDF at 5.5 reads about 0.95, and $P(\text{mass} > 5.5) = 1 - F(5.5)$. Any estimate from about 0.03 to 0.08 is fine.

Why. The CDF gives $P(\text{mass} \leq x)$; the probability above x is what's left over.

Watch for. Reporting $F(5.5) \approx 0.95$ itself as the answer (forgetting to subtract from 1).

Part 3: Standardize and look it up

3a.

Compute the z-score for $x = 5.5$ kg.

Answer. $z = (5.5 - 4.5)/0.6 = 1.67$ (1.667 unrounded).

Why. Subtract the mean, divide by the standard deviation.

Watch for. Dividing by the variance (0.36) instead of the standard deviation. Computing $(4.5 - 5.5)/0.6$, which gives the right size with the wrong sign.

3b.

Look up $\Phi(z)$ for the z-score you just computed (round z to the nearest 0.01 if needed), then use it to compute $P(\text{mass} > 5.5)$. How close was this to your graph-reading estimate from 2a?

Answer. $\Phi(1.67) = 0.9525$, so $P(\text{mass} > 5.5) = 1 - 0.9525 = 0.0475$, about 4.7%. It should be close to the 2a estimate.

```
1 - pnorm(1.67)           # table value
```

```
## [1] 0.04745968
```

```
1 - pnorm(5.5, mu, sigma) # unrounded z, for comparison
```

```
## [1] 0.04779035
```

Why. The table gives the area below z ; the probability above is 1 minus that. The small difference from the unrounded answer (0.0478) comes from rounding z to 1.67.

Watch for. Reporting $\Phi(1.67)$ itself. Looking up the wrong cell (row 1.6, column .07 is the right one).

3c.

Search the table for the value closest to 0.90 and read off the z -score (90% quantile). Convert that z -score back into a body mass.

Answer. The closest table value is $\Phi(1.28) = 0.8997$, so $z \approx 1.28$, and $\text{mass} = 4.5 + 1.28 \times 0.6 \approx 5.27$ kg. About 10% of adults are expected to weigh more than this.

```
pnorm(c(1.28, 1.29))      # 0.8997 is closer to 0.90 than 0.9015
```

```
## [1] 0.8997274 0.9014747
```

```
mu + 1.28 * sigma
```

```
## [1] 5.268
```

```
qnorm(0.9, mu, sigma)     # exact, for comparison
```

```
## [1] 5.268931
```

Why. Reversing the z -score formula: $x = \mu + z\sigma$.

Watch for. Reporting $z = 1.28$ as the answer without converting back to kg. Picking 1.29 (0.9015 is a little farther from 0.90). Using $x = (z - \mu)/\sigma$ or $\mu - z\sigma$.

Group discussion and revision

Compare your Part 3 answers with your group. One thing worth discussing: would this z -table give the right answer if you standardized Act 2's binomial detection counts the same way? Record any changes, new insights, and muddiest points.

Discussion points. No, not reliably. Standardizing, $z = (x - \mu)/\sigma$, works for any distribution with a mean and standard deviation, but it only produces the standard normal curve when the variable is normal to begin with. Act 2's counts are discrete and right-skewed (for $N = 8$, $p = 0.3$), and standardizing changes their location and scale, not their shape, so the table would give the wrong probabilities.

Watch for.

- “It works because the distribution is symmetric.” Symmetry isn't enough: a uniform distribution is symmetric, but its standardized shape is flat, not bell-shaped.
- “It's fine because we're far from 0.” That's about whether a normal model is reasonable for these data in the first place, not about whether standardizing a non-normal variable gives normal probabilities.